

Christian Sassi

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EDUCATION

Master's Degree in Computer Science, Cybersecurity - University of Trento

Graduated with a final grade of 110/110 cum laude.

Trento, Italy

Sep. 2023 – Dec. 2025

- Thesis: **Federated Learning**-based anomaly detection for **Industrial Control Systems (ICS)**.

Bachelor's Degree in Computer Engineering - University of Trento

Graduated with a final grade of 108/110.

Trento, Italy

Sep. 2020 – Sep. 2023

EXPERIENCE

Bug Bounty Program Operator

Huawei

- Conduct **mobile security** research on private **bug bounty programs**.

Feb. 2026 – Present

Munich, Germany

Software Developer

Energie3

- Designed a secure **Python** backend that orchestrates authenticated access to multiple external databases through an **encrypted local metadata layer**.
- Built a web-based **SQL client (HTML, CSS, JavaScript)** for safe query execution with efficient **connection management** and no external dependencies.

Nov. 2025 – Jan. 2026

Reggio Emilia, Italy

Researcher Intern - Cybersecurity

Fondazione Bruno Kessler (FBK)

- Engineered a **federated learning** system that decentralizes the **DAICS** deep-learning framework for anomaly detection in **Industrial Control Systems (ICS)**, training across **7 clients** that share only model updates, never raw data.
- Integrated **adaptive client selection** and workload allocation from **FLAD**, and stabilized distributed training with loss and **gradient clipping** to fix a gradient explosion that collapsed early runs.
- Evaluated on the **SWaT** dataset (**51 sensors/actuators**, 947K records, 36 attack scenarios), reaching **performance comparable to the centralized baseline** while preserving **data privacy**.

Mar. 2025 – Dec. 2025

Trento, Italy

Software Engineer Intern - Embedded Systems

University of Trento

- Characterized **SPARK SR1020 Ultra-Wideband (UWB)** radios, measuring packet-reception rate, latency, and data rate (up to **10 Mbps**) across payloads, **FEC** levels, and modulation modes.
- Optimized **C** firmware for real-time **UWB** communication, reducing the transmission time slot by **50%**.
- Developed **Python** tools to instrument the firmware and automate performance analysis.

Feb. 2023 – Jun. 2024

Trento, Italy

PROJECTS

MITRE Embedded Capture The Flag (eCTF) Competition | C

- Competed in MITRE's **eCTF** national **firmware security** competition as part of a **12-person team**, placing **13th of 116**.
- Wrote defensive **C** code in the embedded firmware: **memory zeroing**, **overflow checks**, and overflow-safe primitives.

Security Testing Analysis for an E-commerce Platform | Java

- Ran **static (SAST)** and **dynamic (DAST)** analysis of a **Java** platform with **SpotBugs** and **OWASP ZAP**.
- Identified **15+ vulnerabilities** including **SQLi**, **XSS**, and **CSRF**, triaging findings for remediation.
- Designed and applied **15+ targeted fixes** for the discovered issues.

T.R.U.S.T. - Secure Chat Application | Rust

- Built an **end-to-end encrypted** chat application in **Rust** (TUI), implementing **Signal** protocol primitives.
- Implemented **X3DH** key establishment, the **Double Ratchet** (forward secrecy), and authenticated **AES-GCM** encryption.

PUBLICATIONS

Real-time Musical Haptics with Ultra-wideband: A Study on Latency, Reliability, and Perception

L. Turchet, C. Sassi, D. Vecchia, G. Picco - Published in IEEE Transactions on Haptics on January 6, 2025.

SKILLS

Spoken Languages: Italian, English

Programming Languages: Python, C, C++, Java, Rust, JavaScript, HTML, CSS, Bash, Solidity, SQL

Security: Vulnerability Assessment, Web Security, Mobile Security, Network Security, Reverse Engineering, Binary Exploitation (Pwn), Injection Attacks, Cryptography, Federated Learning

Security Testing: SAST, DAST, OWASP Top 10, CVE, CWE